

Code-along 08

FirstName LastName

Setup

Packages

Load the standard packages and install `moderndrive`

```
# install.packages("moderndrive")

library(here)
library(tidyverse)
library(haven) # not core tidyverse
library(summarytools)
library(gtsummary)
library(moderndrive)
```

Create an example dataframe

```
# Create columns
literacy <- c(31.7, 79.3, 85.4, 41.0, 54.7, 97.9, 92.8, 67.1, 97.8, 60.4, 87.5, 99.1)
educ <- c(3.74, 10.4, 20.4, 4.46, 9.53, 78.6, 31.5, 16.2, 65.8, 10.3, 30.8, 40.1)

# Combine into a data frame
df <- data.frame(literacy, educ)

# View the data frame
print(df)
```

	literacy	educ
1	31.7	3.74
2	79.3	10.40
3	85.4	20.40
4	41.0	4.46
5	54.7	9.53
6	97.9	78.60
7	92.8	31.50
8	67.1	16.20
9	97.8	65.80
10	60.4	10.30
11	87.5	30.80
12	99.1	40.10

Look at descriptive statistics

`moderndive::tidy_summary()` returns commonly used summary statistics in a tidy format.

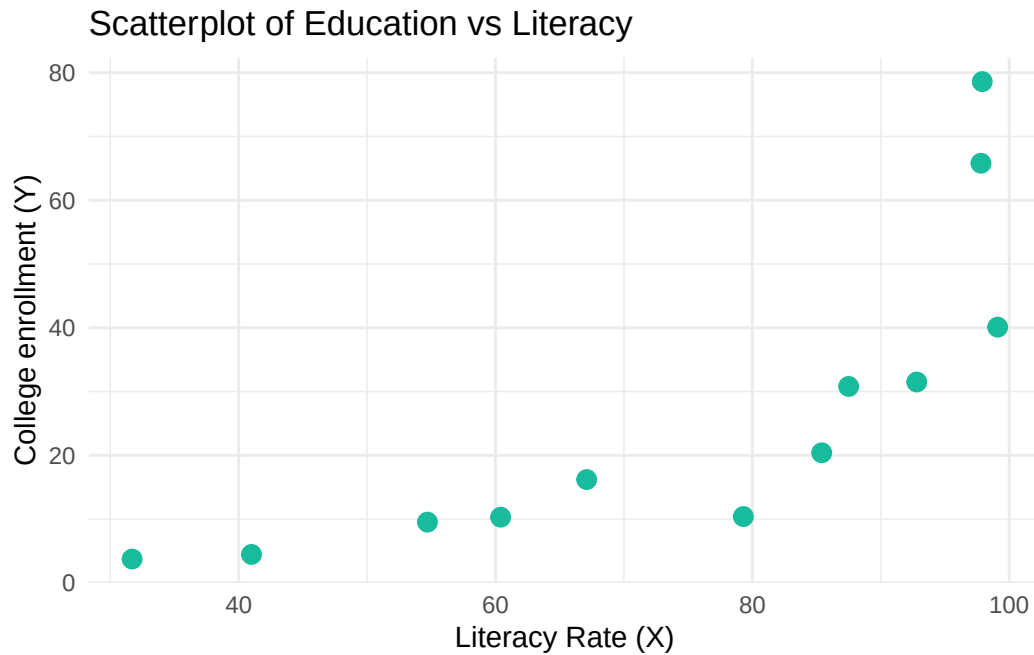
```
df |>
  tidy_summary()
```

A tibble: 2 x 11

	column	n	group	type	min	Q1	mean	median	Q3	max	sd
	<chr>	<int>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	literacy	12	<NA>	numeric	31.7	59.0	74.6	82.4	94.0	99.1	23.2
2	educ	12	<NA>	numeric	3.74	10.1	26.8	18.3	33.6	78.6	24.2

Create a scatterplot.

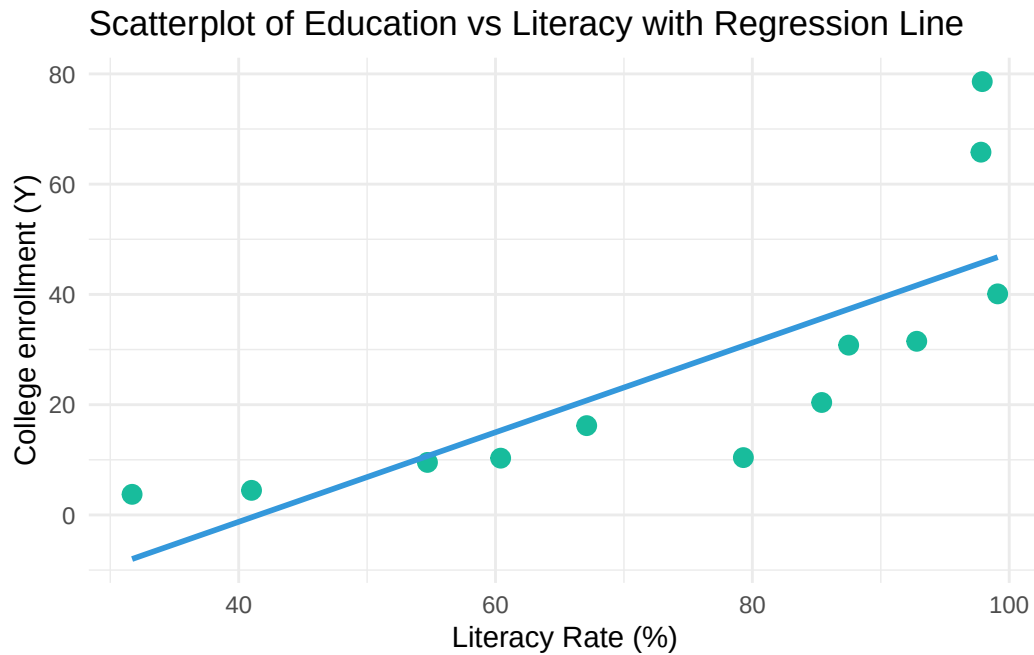
```
# Create scatterplot
df |>
  ggplot(aes(x = literacy, y = educ)) +
  geom_point(color = "#18BC9C", size = 3) +
  labs(
    title = "Scatterplot of Education vs Literacy",
    x = "Literacy Rate (X)",
    y = "College enrollment (Y)"
  ) +
  theme_minimal()
```



Add a regression line to the scatterplot.

```
# Create scatterplot with regression line
df |>
  ggplot(aes(x = literacy, y = educ)) +
  geom_point(color = "#18BC9C", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "#3498DB") +
  labs(
    title = "Scatterplot of Education vs Literacy with Regression Line",
    x = "Literacy Rate (%)",
    y = "College enrollment (Y)"
  ) +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'



Regression

Use base R to run a regression.

```
lm(educ ~ literacy, data = df)
```

Call:

```
lm(formula = educ ~ literacy, data = df)
```

Coefficients:

(Intercept)	literacy
-33.7449	0.8123

Run a correlation test.

```
cor.test(  
  ~ educ + literacy,  
  data = df,  
  method = "pearson",
```

```
na.action = na.omit
)
```

Pearson's product-moment correlation

```
data: educ and literacy
t = 3.942, df = 10, p-value = 0.002766
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.3731885 0.9352545
sample estimates:
      cor
0.7800287
```

Use `get_regression_table()` to run a regression.

```
model <- lm(educ ~ literacy, data = df)

get_regression_table(model)
```

```
# A tibble: 2 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept -33.7      16.0     -2.10   0.062   -69.5    1.98
2 literacy   0.812     0.206     3.94   0.003    0.353    1.27
```

Categorical IV

Add a categorical variable to the example df.

```
# Use mutate and case_when to add the country_group column
df <- df %>%
  mutate(country_group = case_when(
    literacy < 50 ~ "Low Literacy",
    literacy >= 50 & literacy <= 80 ~ "Medium Literacy",
    literacy > 80 ~ "High Literacy"
  ))

head(df)
```

	literacy	educ	country_group
1	31.7	3.74	Low Literacy
2	79.3	10.40	Medium Literacy
3	85.4	20.40	High Literacy
4	41.0	4.46	Low Literacy
5	54.7	9.53	Medium Literacy
6	97.9	78.60	High Literacy

Use `get_regression_table()` with a categorical IV.

```
model_2 <- lm(educ ~ country_group, data = df)

get_regression_table(model_2)
```

```
# A tibble: 3 x 7
  term                                estimate std_error statistic p_value lower_ci upper_ci
  <chr>                                <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept                           44.5      6.94      6.42     0      28.8    60.2
2 country_group: Low Lit~             -40.4     13.9     -2.91   0.017   -71.8   -9.03
3 country_group: Medium ~             -32.9     11.0      -3      0.015   -57.8   -8.10
```

R^2

Use `get_regression_points()` to see the residuals.

```
model <- lm(educ ~ literacy, data = df)

get_regression_points(model)
```

```
# A tibble: 12 x 5
  ID   educ literacy educ_hat residual
  <int> <dbl>   <dbl>   <dbl>   <dbl>
1     1  3.74    31.7    -8.00    11.7
2     2 10.4    79.3    30.7    -20.3
3     3 20.4    85.4    35.6    -15.2
4     4  4.46    41     -0.44     4.9
5     5  9.53    54.7    10.7    -1.16
6     6 78.6    97.9    45.8    32.8
7     7 31.5    92.8    41.6   -10.1
8     8 16.2    67.1    20.8    -4.56
```

9	9	65.8	97.8	45.7	20.1
10	10	10.3	60.4	15.3	-5.02
11	11	30.8	87.5	37.3	-6.53
12	12	40.1	99.1	46.8	-6.65

Calculate R^2

```
my_test <- cor.test(
  ~ educ + literacy,
  data = df,
  method = "pearson",
  na.action = na.omit
)

my_test$estimate
```

```
cor
0.7800287
```

```
(my_test$estimate)^2
```

```
cor
0.6084447
```

Use `get_regression_summaries()` to see R^2 .

```
model <- lm(educ ~ literacy, data = df)

get_regression_summaries(model)
```

```
# A tibble: 1 x 9
  r_squared adj_r_squared   mse  rmse sigma statistic p_value    df  nobs
  <dbl>      <dbl> <dbl> <dbl> <dbl>    <dbl>   <dbl> <dbl> <dbl>
1   0.608      0.569  210.  14.5  15.9     15.5   0.003     1    12
```